187 BORDEN AVENUE NORTH

KITCHENER, ONTARIO

CONCEPTUAL FUNCTIONAL SERVICING & STORMWATER MANAGEMENT REPORT

Self-Directed Portfolio Project

Prepared for technical portfolio demonstration

August 2026

Conceptual design only - not for municipal approval or construction.

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1 Background

This Conceptual Functional Servicing and Stormwater Management Report has been prepared for a proposed development at 187 Borden Avenue North in the City of Kitchener, Ontario**.** The purpose of this report is to evaluate the conceptual servicing and storm drainage requirements associated with the redevelopment of the subject property and to develop an integrated grading, drainage, and stormwater management strategy.

The assessment considers existing and proposed drainage conditions, changes in runoff characteristics resulting from redevelopment, stormwater quantity control, stormwater quality treatment, on-site detention storage, outlet control, the minor storm sewer system, and the major overland drainage system.

Autodesk Civil 3D was used to develop the conceptual site model, including existing and proposed surfaces, grading, alignments, profiles, corridor elements, feature lines, surface breaklines, drainage features, and the storm sewer pipe network. ArcGIS was used to prepare the site location mapping and provide geographic context for the subject property.

This report has been prepared as a self-directed conceptual portfolio project for technical demonstration purposes. It is not intended for municipal approval or construction. Detailed survey information, utility verification, geotechnical information, downstream capacity assessment, confirmation of applicable agency criteria, and professional engineering review would be required before advancing the design to detailed design or construction.

1.1 Site Location

The subject property is located at 187 Borden Avenue North, at the intersection of Borden Avenue North and East Avenue in the City of Kitchener, Ontario. The property is situated within the King Street East Neighbourhood Secondary Plan Area and has a total area of approximately 0.115 ha. The site location and surrounding municipal road network are illustrated in Figure 1.

Figure 1 Site Location Plan

1.2 Existing Site Conditions

Under existing conditions, the subject property contains two brick residential dwellings with associated driveways and partially hardscaped areas. The remainder of the property generally consists of pervious landscaped areas.

Existing municipal infrastructure surrounding the subject property includes water, sanitary sewer, and storm sewer services within Borden Avenue North and East Avenue. A summary of the existing municipal services adjacent to the subject property is provided in Table 1.

Table 1 Existing Municipal Services

Service	Size	Location
Water	300 mm (PVC)	East Ave
Water	150 mm (CI)	Borden Ave North
Storm	250 mm (Conc)	East Ave
Storm	250 mm (Conc)	Borden Ave North
Sanitary	250 mm (Conc)	East Ave
Sanitary	200 mm (VCP)	Borden Ave North

2 Proposed Development

The proposed development consists of three-storey stacked townhouse units, with a primary vehicular access from Borden Avenue North and an associated on-site parking area. The development will also include associated site servicing, grading, drainage, and stormwater management infrastructure.

The proposed site servicing and stormwater management strategy has been developed conceptually to accommodate the proposed redevelopment while providing appropriate drainage, quantity control, quality treatment, and conveyance of stormwater runoff.

3 Proposed Grading

A conceptual finished-grade surface was developed for the proposed development to provide positive drainage away from the proposed buildings and direct stormwater runoff toward the proposed catchbasins and designated surface drainage routes.

The internal paved areas are graded toward the proposed catchbasins to collect runoff generated within the site. Localized low points are provided at selected catchbasins to permit temporary surface ponding during major storm events. The maximum design ponding depth is limited to approximately 0.20 m above the applicable catchbasin rim elevation.

Defined ridge and valley lines were incorporated into the finished-grade surface to establish drainage divides and direct runoff toward the intended catchbasins. Surface drainage behaviour was reviewed using Civil 3D Water Drop analysis and user-defined contour analysis to confirm the direction of surface runoff and assess the extent of localized ponding.

The proposed grading, catchbasin locations, drainage areas, ponding limits, and overland flow routes are illustrated on Figure 3 Post-Development Drainage and Stormwater Management Plan in Appendix A**.**

4 Stormwater Management

Redevelopment of the site increases impervious coverage and consequently increases the peak stormwater runoff generated from the property. The conceptual stormwater management strategy has therefore been developed to address both quantity control and quality control prior to discharge to the municipal storm sewer system. The proposed stormwater management system uses a combination of surface ponding, available storage within the private storm sewer system and drainage structures, and an outlet control device to attenuate post-development peak flows. Stormwater quality treatment is proposed through an oil-grit separator (OGS) located upstream of the connection to the municipal storm sewer.

4.1 Stormwater Management Criteria

The following conceptual stormwater management criteria have been adopted for this portfolio assessment:

Quantity control	Control the post-development 100-year storm runoff to the pre-development 5-year peak discharge rate.
Hydrology	Use the Rational Method for peak flow calculations and the Modified Rational Method for conceptual detention-storage assessment.
Rainfall	Use the City of Kitchener IDF parameters contained in the project calculation workbook.
Time of concentration	Adopt Tc = 10 minutes for the conceptual small-site assessment.
Surface storage	Provide controlled shallow ponding with a conceptual maximum depth of approximately 0.20 m.
Quality control	Provide treatment through an oil-grit separator (OGS) or equivalent approved treatment system before discharge to the municipal storm sewer.
Drainage	Provide coordinated minor storm sewer and major overland drainage systems.

Quantity-control objective: $Q_{post,100\text{-year}} \leq Q_{pre,5\text{-year}}$

The resulting conceptual drainage sequence is Site Runoff → Catchbasins → Storm Sewer Network → On-Site Storage → Flow Control → Quality Treatment → Municipal Storm Sewer.

4.2 Quantity Control

The quantity-control objective is to mitigate the increase in runoff resulting from redevelopment by temporarily storing excess runoff on site and releasing it through a controlled outlet. The governing criterion for this conceptual assessment is to control the post-development 100-year storm discharge to the pre-development 5-year peak discharge rate.

4.2.1 Pre- and Post-Development Runoff Coefficients

Table 2 Summary of Pre- and Post-Development Runoff Coefficients

Condition	Area (ha)	Weighted Runoff Coefficient
Pre-Development	0.115	0.46
Post-Development	0.115	0.78

The increase in weighted runoff coefficient represents the increased runoff response associated with redevelopment. Detailed land-use calculations are provided in Table 5 and Table 6 in Appendix B.

4.2.2 Pre- and Post-Development Flow Calculations

Peak runoff rates were calculated using the Rational Method:

$$Q = 2.78\,C\,I\,A$$

where Q is the peak runoff rate (L/s), C is the weighted runoff coefficient, I is the rainfall intensity (mm/hr), and A is the drainage area (ha). Rainfall intensity was calculated from the City of Kitchener IDF parameters contained in the project workbook. A time of concentration of 10 minutes was adopted.

Table 3 summary of Pre & Post Development Flows

Return Period	Pre Development (liters/sec)	Post Development (liters/sec)
Q5	16.27743	27.42529
Q100	29.16834	49.14474

Allowable quantity-control release: $Q_{Allowable} = Q_{pre,5\text{-}year} = 16.28$ L/s

The uncontrolled post-development 100-year peak flow is approximately 49.14 L/s. The conceptual stormwater management system is therefore intended to reduce the 100-year post-development discharge from approximately 49.14 L/s to no more than the 5-year pre-development allowable release rate of 16.28 L/s. Detailed Rational Method calculations are provided in Table 8 and Table 9 in Appendix B.

4.2.3 Orifice Control

To ensure that only allowable 5-year predevelopment flow is released from the project area, a 60 mm diameter orifice pipe (rated flow 15.35 l/sec) is proposed to be installed downstream of storm manhole MH2. The orifice sizing calculations are attached as Table 10, Appendix B.

For this conceptual portfolio project, a 60 mm diameter opening has been adopted as a practical trial design constraint to reduce susceptibility to blockage. Minimum orifice dia 75 mm as per the MECP but that opening would discharge approximately 23.96 L/s, which is greater than the allowable pre-development 5-year release rate of 16.28 L/s. Accordingly, the final outlet-control arrangement requires further iteration rather than simply adopting the theoretical 60 mm opening.

4.2.4 On-Site Storage

The Modified Rational Method was used to estimate the preliminary detention-storage requirement. A range of storm durations was evaluated for the post-development 100-year storm. For each duration, the cumulative runoff volume was compared with the cumulative controlled outflow volume. The maximum onsite storage requirement is 25.16 m3 for 100yr storm in order to control the site flows to 15.35 l/sec (orifice control). The calculation of required onsite storage volume is given in Table 11, Appendix B

The required Onsite storage is provided as follows;

Table 4 Available On-Site Storage Summary

Storage Component	Available Storage (m^3)
Surface Ponding	17.84
Storm Sewer Pipes	3.70
Drainage Structures	7.02
Total Available Storage	28.02

The project workbook evaluates the storage requirement using the 60 mm trial orifice discharge of approximately 15.35 L/s. The analysis identifies a critical duration of approximately 25 minutes and a maximum preliminary storage requirement of approximately 25.16 m^3.

Preliminary storage balance: 28.02 m^3 available > 25.16 m^3 required

4.2.5 Surface Ponding Storage

The proposed grading incorporates three controlled shallow ponding areas. The conceptual maximum ponding depth is approximately 0.20 m, with a corresponding high-water elevation of approximately 335.71 m in the project storage calculation.

The preliminary surface-storage volume is approximately 17.84 m^3. The ponding limits are shown on in Appendix A. For detailed design, surface storage should be calculated from the three-dimensional stage-storage relationship of the final proposed surface rather than by assuming the maximum ponding area is uniformly at the maximum depth.

4.2.6 Storm Sewer Pipe Storage

Storage within the proposed 300 mm diameter storm sewer pipes contributes to the available detention volume when the system is surcharged. The project calculation includes approximately

52.4 m of 300 mm diameter pipe, providing approximately 3.70 m^3 of full-pipe storage. Detailed pipe storage calculations is provided in Table 12 in Appendix B.

4.2.7 Drainage Structure Storage

Storage within three drainage structures was estimated using conceptual internal plan dimensions of approximately 1.2 m by 1.2 m and the effective storage heights contained in the project workbook. The resulting structure-storage contribution is approximately 7.03 m^3. Detailed structure storage calculations is provided in Table 12 in Appendix B.

4.3 Quality Control

Stormwater runoff from proposed asphalt, concrete, and other developed surfaces may contain sediment, suspended solids, hydrocarbons, and associated pollutants. An oil-grit separator (OGS) is conceptually proposed downstream of the site storm sewer network and upstream of the municipal storm sewer connection to provide stormwater quality treatment.

The final OGS model and size would be selected during detailed design based on the tributary drainage area, percentage imperviousness, applicable treatment objective, treatment flow rate, hydraulic conditions, inlet and outlet elevations, and verified treatment performance. No proprietary OGS unit is considered finalized as part of this conceptual portfolio project.

4.4 Minor Storm Sewer Network

The minor drainage system consists of proposed catchbasins, drainage structures, and storm sewer pipes used to collect and convey runoff toward the downstream outlet and municipal connection. The conceptual network was developed using Civil 3D Pipe Network tools and coordinated with the proposed finished-grade surface.

The storm sewer design considered catchbasin locations, rim elevations, pipe invert elevations, pipe diameters, pipe slopes, available cover, structure depths, and the downstream connection condition. Parts lists, network rules, pipe and structure styles, and labels were used to develop and document the system. The network was reviewed in both plan and profile.

The proposed storm sewer network is shown on Figure 3 Post-Development Drainage and Stormwater Management Plan, and the corresponding pipe-network profile is shown on Figure 4 in Appendix A.

4.5 Major System Drainage

The major drainage system provides a defined overland drainage route when runoff exceeds the capacity of the minor storm sewer system or cannot immediately enter the catchbasins. The proposed finished-grade surface was developed to direct runoff toward designated low points.

The major overland drainage route is shown on Figure 3 in Appendix A. Detailed design would require confirmation of emergency overflow routes and verification that major flows can be conveyed without unacceptable impacts to buildings, adjacent properties, or public infrastructure.

5 Conclusions

A conceptual grading, drainage, and stormwater management strategy has been developed for the proposed redevelopment at 187 Borden Avenue North, Kitchener, Ontario. The conceptual drainage area is approximately 0.115 ha. The weighted runoff coefficient increases from approximately 0.46 under pre-development conditions to approximately 0.78 under post-development conditions.

The 5-year pre-development peak runoff is approximately 16.28 L/s and has been adopted as the allowable quantity-control release rate. The uncontrolled 100-year post-development peak runoff is approximately 49.14 L/s. The governing quantity-control objective is therefore to control the post-development 100-year discharge to the pre-development 5-year release rate of 16.28 L/s.

The 60 mm theoretical trial outlet discharges approximately 15.35 L/s and produces a preliminary Modified Rational Method storage requirement of approximately 25.16 m^3.

The conceptual site provides approximately 17.84 m^3 of surface ponding storage, 3.70 m^3 of storm sewer storage, and 7.03 m^3 of drainage-structure storage, for a total available storage of approximately 28.02 m^3.

The preliminary storage balance therefore provides approximately 2.86 m^3 of storage above the requirement calculated using the 60 mm theoretical trial outlet.

A 75 mm trial orifice has been adopted as a practical conceptual design constraint; under the current assumed head it would discharge approximately 23.96 L/s and would not satisfy the 16.28 L/s allowable pre-development 5-year release rate.

Stormwater quality treatment is conceptually provided through an OGS upstream of the municipal storm sewer connection.

This report and the associated drawings have been prepared as a self-directed conceptual portfolio exercise and are not intended for construction or municipal approval. Detailed design would require field survey, current municipal and agency criteria confirmation, utility verification, downstream capacity assessment, final hydraulic routing, final OGS selection and sizing, and professional engineering review.

APPENDIX A - DRAWINGS

Figure 2 Drainage Plan 1 - Pre-Development

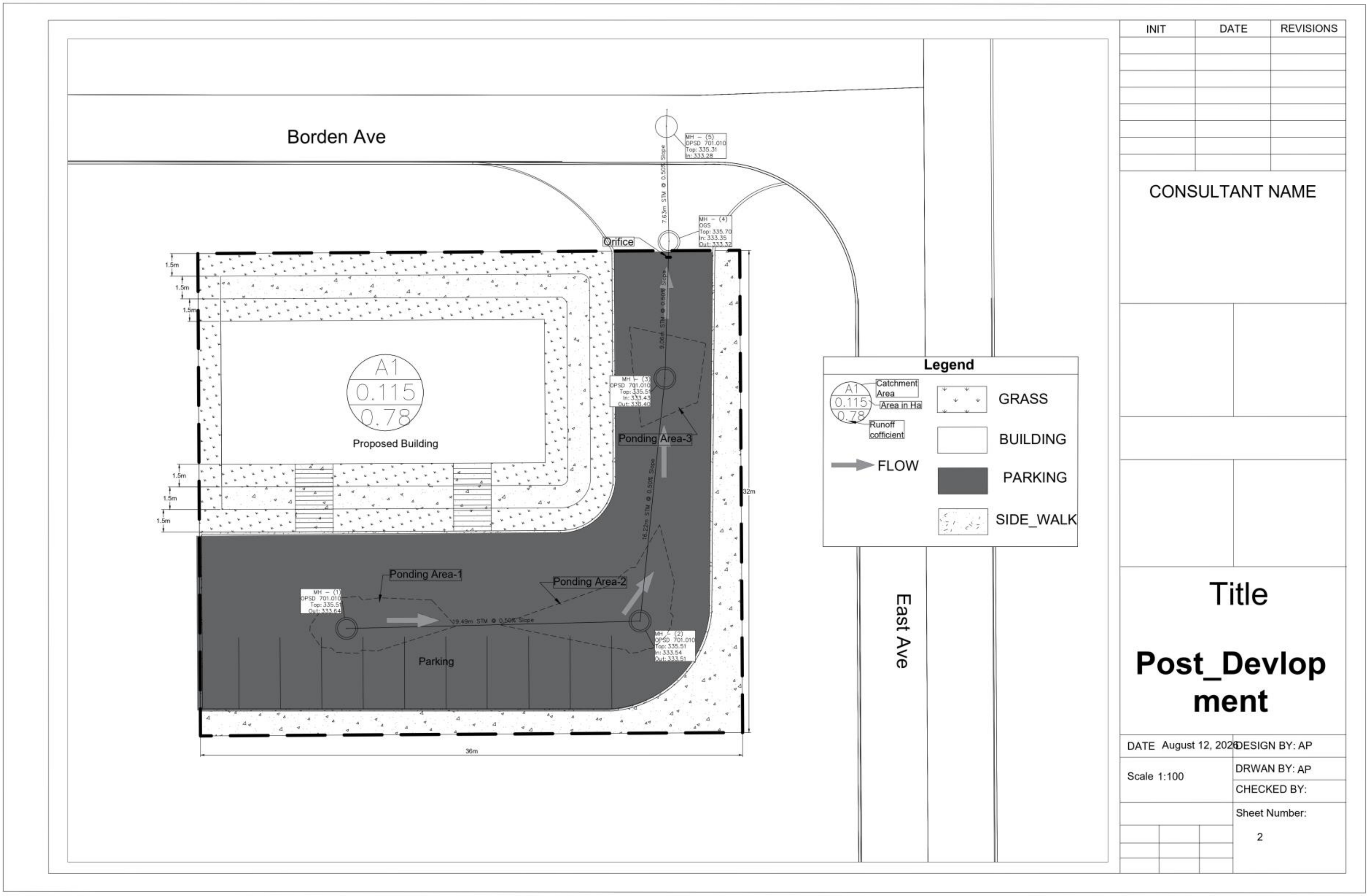

Figure 3 Post-Development Drainage and Stormwater Management Plan

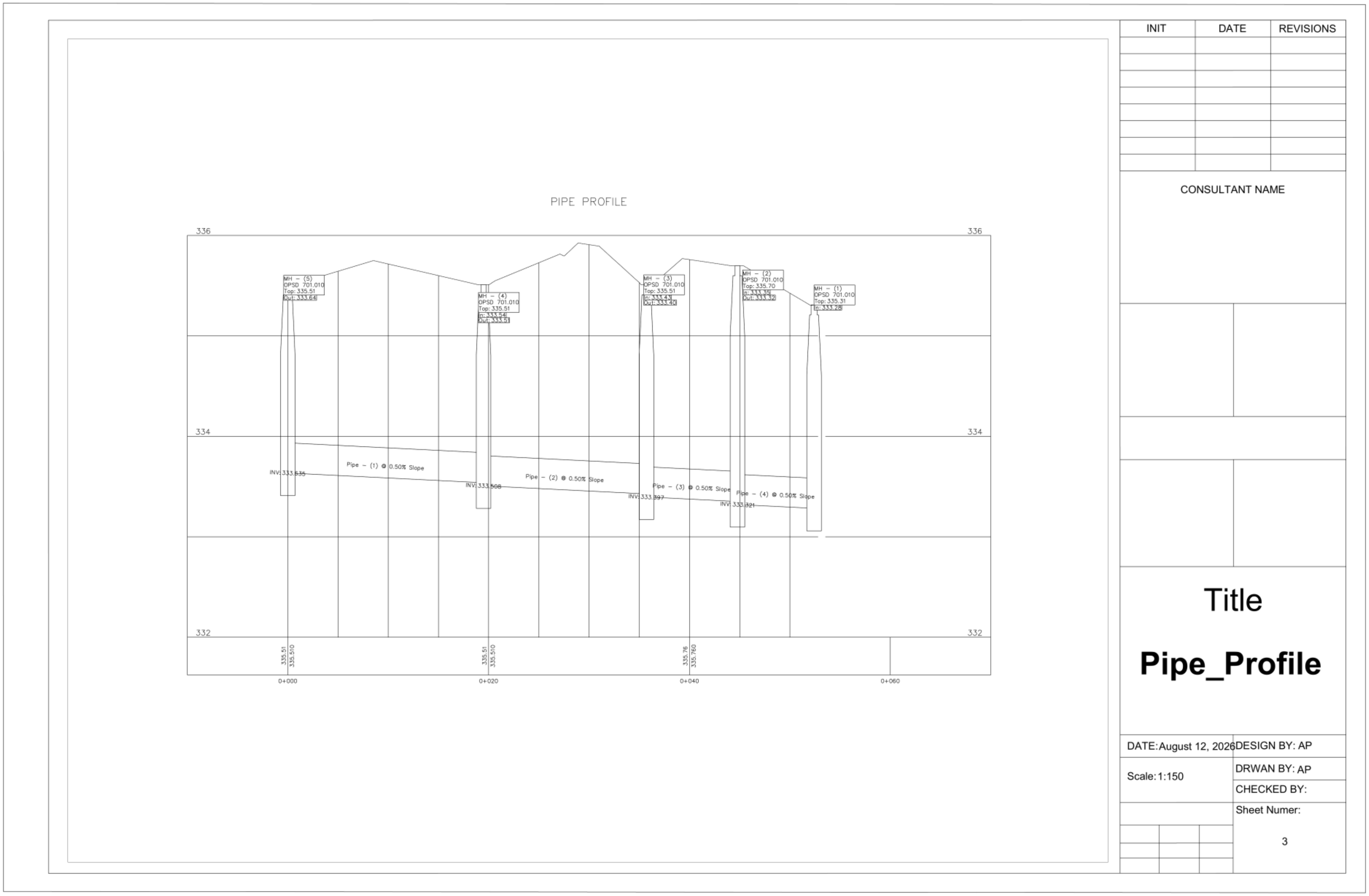

Figure 4 Storm Sewer Profile

APPENDIX B - DESIGN CALCULATIONS

Table 5 Existing Land Use and Composite Runoff Coefficient

Land Use	Area (m²)	Runoff Coefficient, C
Building	334.117	0.90
Grass	716.989	0.20
Asphalt	100.844	0.90
Total / Weighted	1,152	0.4639

Table 6 Proposed Land Use and Composite Runoff Coefficient

Land Use	Area (m²)	Runoff Coefficient, C
Building	205.035	0.90
Grass	195.119	0.20
Concrete	247.219	0.90
Asphalt	504.627	0.90
Total / Weighted	1,152	0.7814

Table 7 Intensity-Duration-Frequency Parameters, City of Kitchener

RETURN	PERAMETERS			
Years	A	B	C	Intensity (mm/hr)
2	743	6	0.7989	81.10
5	1593	11	0.8789	109.68
10	2221	12	0.908	134.16
25	3158	15	0.9355	155.47
50	3886	16	0.9495	176.19
100	4688	17	0.9624	196.54

Post-Development Peak Flow Calculations

Table 8 Peak Flows - Existing Condition

<table>
<tr><th colspan="2" rowspan="2">Existing_Condition</th><th>Area (ha)</th><th>Runoff Coefficient</th><th>Tc (min)</th><th></th></tr>
<tr><td>0.115083</td><td>0.463887</td><td>10</td><td></td></tr>
<tr><td colspan="6">Storm Event</td></tr>
<tr><td>2 Year</td><td>5 Year</td><td>10 Year</td><td>25 Year</td><td>50 Year</td><td>100 Year</td></tr>
<tr><td colspan="6">Peak Flow (cms)</td></tr>
<tr><td>0.0120</td><td>0.0163</td><td>0.0199</td><td>0.0231</td><td>0.0261</td><td>0.0292</td></tr>
</table>

Table 9 Peak Flows - Proposed Condition

Proposed_Condition		Area (ha)	Runoff Coefficient	Tc (min)	
		0.115118	0.781353	10	
Storm Event					
2 Year	5 Year	10 Year	25 Year	50 Year	100 Year
Peak Flow (cms)					
0.0203	0.0274	0.0335	0.0389	0.0441	0.0491

Orifice Calculation

$$Q = C_d A \sqrt{2gh}$$

where $\mathbf{C_d}$= 0.8

Table 10 Orifice Control Calculation Summary

Manhole	**HWL (m)**	**Orifice Inv (m)**	$\mathbf{C_d}$	**A (m^2)**	**g (m/s^2)**	**Orifice dia (m)**	**h (m)**	**Q**	
								(m^3/s)	(L/s)
MH2	335.71	333.32	0.8	0.02826	9.81	0.06	2.35	0.01535	15.35

Proposed Orifice pipe of Dia. 60 mm with flow control of 15.35 lit/sec

Table 11 Modified Rational Method Storage Calculation (60 mm Orifice Outlet)

Duration (min)	**Intensity (mm/hr)**	**Inflow (m^3/s)**	**Runoff Volume (m^3)**	**Outflow Volume (m^3)**	**Storage (m^3)**
10	196.54	0.049145	29.49	9.21	20.28
15	166.89	0.041732	37.56	13.82	23.74
20	145.13	0.036290	43.55	18.42	25.13
25	128.46	0.032122	48.18	23.03	25.16
30	115.28	0.028827	51.89	27.63	24.26
35	104.59	0.026154	54.92	32.24	22.69
40	95.75	0.023942	57.46	36.84	20.62
45	88.31	0.022081	59.62	41.45	18.17
50	81.95	0.020493	61.48	46.05	15.43
55	76.47	0.019122	63.10	50.66	12.44
60	71.69	0.017925	64.53	55.27	9.27
65	67.47	0.016872	65.80	59.87	5.93
70	63.74	0.015938	66.94	64.48	2.46
75	60.40	0.015103	67.97	69.08	-1.12

Storage Calculations

Table 12 Available On-Site Storage Calculation Drainage Structure, Pipes, Parking

Drainage Structures

Structure	Length (m)	Width (m)	Effective Height (m)	Volume (m^3)
MH(1)	1.20	1.20	1.950	2.204
MH(2)	1.20	1.20	2.077	2.348
MH(3)	1.20	1.20	2.188	2.473
			Total	7.025

Storm Sewer Pipes

Length (m)	Diameter (m)	Volume (m^3)
19.5	0.30	1.378
16.2	0.30	1.145
9.1	0.30	0.643
	Total	3.165

Surface Ponding

Ponding Area	Area (m^2)	Depth (m)	Volume (m^3)	HWL (m)
1	26.178	0.20	5.236	335.71
2	39.405	0.20	7.881	335.71
3	23.593	0.20	4.719	335.71
		Total	17.835	

TOTAL AVAILABLE STORAGE = 28.02 m^3